Bocconi

COMPUTER SCIENCE

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Lesson 9

What-if analysis: solving financing and investment problems

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Objectives of the lesson

- Understand the meaning of what-if analysis and learn about the tools available in the spreadsheet
- Learn how to use Excel's financial functions to solve financing and investment problems

What-if analysis

- What-if analysis is a decision-making technique used to evaluate the effects of different scenarios on an outcome by changing key variables
- It helps in assessing risks, predicting future results, and making informed choices by exploring various possibilities
- What-if analysis tools in Excel include:
  - Goal Seek, which allows you to find what value a cell must contain in order to obtain a specific value as a result of the calculation in another cell
  - Scenario Manager, which allows you to examine different possible outcomes (of one or more cells) by adjusting the input values in other cells

Financial functions

- They allow to make calculations that are typical of financial math and corporate finance, without manually setting up complex calculations and formulas
- They allow to calculate installments, interest rates, payments, amortizations and to evaluate investments, loans etc.
- Examples: PMT, PPMT, IPMT, FV

Our case

- One of our clients won the first prize in a lottery worth €300,000
- The amount will be paid in 10 years with constant monthly installments of €2,500
- Since our client's monthly income already allows him to meet all his needs, he has asked us to help him evaluate how to make use of this amount without changing his habits in any way
- The options he would like to consider are:
  - Buying a €260,000 house with a 10-year fixed-rate **mortgage**
  - A ten-year fixed-rate **savings plan**

Our case: mortgage option

- If we consider investing the winnings in a €260,000 house, we must assess whether the monthly installment of the ten-year fixed-rate mortgage proposed by the bank (with payment of each installment at the end of the month) is compatible with the new monthly budget of €2,500

- If the mortgage payment is higher, we could consider negotiating a lower interest rate with the bank. It is therefore necessary to calculate what percentage of interest would allow for a monthly installment equal to the monthly availability of €2,500

- The conditions proposed by the bank, however, follow the classic French amortization model, i.e. the method of repaying the loan in which the installments are constant over time, but the composition of the installment changes progressively

The French amortization model

- In the **French amortization** model each installment is therefore made up of two parts:
  - **Interest amount**: this is calculated on the residual debt (i.e. on the amount still to be repaid). At the beginning it is high, because the residual debt is greater
  - **Principal amount**: this is the part of the installment that actually reduces the outstanding debt. It is lower at first, but increases over time
- The main features of this type of fixed-rate mortgage are therefore:
  - Fixed installments: you pay the same amount every month
  - More interest at the beginning: in the early years, you pay more interest and less principal
  - More principal towards the end: as the debt is reduced, the interest decreases and the principal amount increases

Our case: mortgage amortization plan

- Using the data in the Mortgage worksheet, we are asked to calculate the mortgage amortization plan, i.e. the series of constant payments to be made over the 10-year period, highlighting both the interest portion and the principal portion

- The Amortization Plan worksheet already contains a table to be completed with all the required calculations

Our case: savings plan option

- Our client is also considering the option of investing his winnings in a savings plan
- He therefore asks us to calculate the amount he will have available after paying €2,500 every month for 10 years
- The conditions offered by the bank are already present in the Investment worksheet and include some alternative options to those specifically requested by the client
- In addition to calculating the amount available at the end of the savings plan, we are asked to calculate the total amount of interest accrued on the €300,000 winnings

Our case: savings plan scenarios

- Finally, we are asked to show the same values in a summary table (amount available at the end of the savings plan and related interest accrued) for each of the other conditions proposed by the bank
- To improve the readability of the data in the summary table, we are asked to assign each cell containing the data in the range from B2 to B6 an alternative name to the references commonly used to identify it

Tools to learn with today's lesson

- Financial functions: PMT, PPMT, IPMT, FV
- Naming cells
- What-if analysis tools: Goal Seek, Scenario Manager

Book references

Excel for students in economics and finance, 2nd edition:

Chapters 3.3.1, 3.3.2, 3.3.3, 3.3.5, 6.1.1, 6.1.3, 6.1.4

Excel workbook, 4th edition:

Exercises on the functions and tools of the lesson